



Commodore C64 — Chip Pinouts

MAJOR SOCKETED ICS · CPU · VIC-II · SID · CIA · PLA · RAM · TEST & REPAIR

DIP pinouts seen from the top, notch up, pin 1 at top-left. See the **Legend** card for signal notation & the colour key. **6567/6569**=NTSC/PAL; **8565/8580**=later 5V/9V HMOS-II parts.

Legend — reading the pinouts notation

/NAMEActive LOW — the signal is true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /NMI /CS /WE /RAS /CAS.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. e.g. A0–A15, D0–D7, BA. R/W is a level: HIGH = read, LOW = write.

Colour key: Vcc / Vdd = power · GND / Vss = ground · ϕ = clock. **Orientation:** chip seen from the top, notch up — pin 1 is top-left, numbers run down the left side then up the right.

6510 CPU

40-DIP

ϕ 0 in	1	40	/RES
RDY	2	39	ϕ 2 out
/IRQ	3	38	R/W
/NMI	4	37	D0
AEC	5	36	D1
Vcc	6	35	D2
A0	7	34	D3
A1	8	33	D4
A2	9	32	D5
A3	10	31	D6
A4	11	30	D7
A5	12	29	P0
A6	13	28	P1
A7	14	27	P2
A8	15	26	P3
A9	16	25	P4
A10	17	24	P5
A11	18	23	A15
A12	19	22	A14
A13	20	21	GND

C64 CPU @ ~1 MHz. Pins 24–29 = I/O port P0–P5: bank-switch /LORAM /HIRAM /CHAREN + Datasette.
Fails: blank/black screen, dead machine.

VIC-II video

40-DIP

DB6	1	40	Vcc
DB5	2	39	DB7
DB4	3	38	DB8
DB3	4	37	DB9
DB2	5	36	DB10
DB1	6	35	DB11
DB0	7	34	A10
/IRQ	8	33	A9
LP	9	32	A8
/CS	10	31	A7
R/W	11	30	A6
BA	12	29	A5
Vdd	13	28	A4
COLOR	14	27	A3
S/LUM	15	26	A2
AEC	16	25	A1
ϕ 0	17	24	A0
/RAS	18	23	A11
/CAS	19	22	ϕ IN
Vss	20	21	ϕ CL

Video + DRAM refresh + CPU clock. **6567**=NTSC, **6569**=PAL (Vdd 13=+12V); **8565/8562**=5V HMOS-II. **Fails:** no/garbled picture, bad colour; runs hot.

6581 SID sound

28-DIP

CAP1A	1	28	Vdd
CAP1B	2	27	AUDIO
CAP2A	3	26	EXT IN
CAP2B	4	25	Vcc
/RES	5	24	POTX
ϕ 2	6	23	POTY
R/W	7	22	D7
/CS	8	21	D6
A0	9	20	D5
A1	10	19	D4
A2	11	18	D3
A3	12	17	D2
A4	13	16	D1
GND	14	15	D0

3-voice synth + 2 paddle A/D (POTX/Y) @ \$D400.
6581: Vdd 28=+12V, Vcc 25=+5V; **8580**=+9V.
Fails: no/weak/distorted sound, dead paddles.

6526 CIA x2

40-DIP

Vss	1	40	CNT
PA0	2	39	SP
PA1	3	38	RS0
PA2	4	37	RS1
PA3	5	36	RS2
PA4	6	35	RS3
PA5	7	34	/RES
PA6	8	33	D0
PA7	9	32	D1
PB0	10	31	D2
PB1	11	30	D3
PB2	12	29	D4
PB3	13	28	D5
PB4	14	27	D6
PB5	15	26	D7
PB6	16	25	ϕ 2
PB7	17	24	/FLAG
/PC	18	23	/CS
TD	19	22	R/W
Vcc	20	21	/IRQ

CIA1 \$DC00: keyboard, joysticks, IRQ. **CIA2** \$DD00: serial bus, user port, NMI. **Fails:** dead keyboard/ports/joystick, no disk.

PLA 82S100

28-DIP

FE	1	28	Vcc
I7	2	27	I8
I6	3	26	I9
I5	4	25	I10
I4	5	24	I11
I3	6	23	I12
I2	7	22	I13
I1	8	21	I14
I0	9	20	I15
F7	10	19	/CE
F6	11	18	F0
F5	12	17	F1
F4	13	16	F2
GND	14	15	F3

Address decode + bank switching: 16 inputs → 8 chip-selects (F0–F7). MOS 906114-01. **The #1 C64 failure** — runs hot. **Fails:** black screen, won't boot. Sub: GAL/PLankton.

4164 DRAM 64Kx1

16-DIP

NC	1	16	Vss
DIN	2	15	/CAS
/WE	3	14	DOUT
/RAS	4	13	A6
A0	5	12	A3
A2	6	11	A4
A1	7	10	A5
Vcc	8	9	A7

Main RAM: 8x = 64KB. A 41256 substitutes if pins 1+16 are linked. **Fails:** garbage/checkerboard screen, missing RAM, random crashes.

2114 colour SRAM

18-DIP

A6	1	18	Vcc
A5	2	17	A7
A4	3	16	A8
A3	4	15	A9
A0	5	14	D1
A1	6	13	D2
A2	7	12	D3
/CS	8	11	D4
GND	9	10	/WE

1Kx4 static RAM = colour RAM at \$D800. **Fails:** wrong / flashing character colours.

ROMs

mask ROM

901226 / U3 BASIC ROM — 2364, 8K, 24-pin

901227 / U4 KERNAL ROM — 2364, 8K, 24-pin

901225 / U5 Character ROM — 2332, 4K, 24-pin

Mask ROMs rarely fail but can. Replace with 2764 / 27128 EPROM via a pin adapter. **Fails:** garbage on boot, missing/garbled characters.

Test & Repair

tips

Power rails +5V logic · +12V (SID/VIC Vdd) · 9VAC in

Black screen PLA → CPU → clock / oscillator

Garbage screen DRAM (swap 4164s) or address logic

No picture VIC-II

No sound SID

Dead keys/disk CIA1 / CIA2

Swap order for a dead board: **PLA → VIC → CPU → RAM → CIA**. A Dead-Test or diagnostic cartridge speeds this up. Socket chips before testing.